

Professional Summary

AI/ML Engineer with 3+ years of experience designing, developing, and deploying scalable machine learning and Generative AI solutions. Proficient in Python, SQL, Scikit-learn, TensorFlow, PyTorch, FastAPI, PySpark, Docker, Kubernetes, MLflow, PostgreSQL, AWS, LangChain, OpenAI APIs, and Retrieval-Augmented Generation (RAG). Experienced in building end-to-end ML pipelines, feature engineering, model training, hyperparameter tuning, model deployment, REST API development, and MLOps practices. Skilled in prompt engineering, vector search, semantic retrieval, model monitoring, and performance optimization to deliver reliable production-ready AI applications. Adept at collaborating with cross-functional teams, solving complex technical challenges, and delivering high-quality AI solutions using Agile methodologies.

Technical Skills

Programming Languages: Python, SQL, R, Bash, Java
Python Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, LightGBM, CatBoost, Matplotlib, Plotly, OpenCV, Hugging Face Transformers, LangChain, MLflow
Machine Learning: Supervised Learning, Unsupervised Learning, Deep Learning, Feature Engineering, Model Training, Model Evaluation, Model Optimization, Predictive Analytics, Time Series Forecasting
Generative AI: OpenAI API, LangChain, RAG, Prompt Engineering, Vector Databases, LLM Fine-Tuning, AI Agents
Data Engineering: PySpark, Apache Spark, Apache Airflow, ETL Pipelines, Data Preprocessing, Data Validation
Frameworks & Deployment: FastAPI, Flask, REST APIs, Docker, Kubernetes, CI/CD, Model Monitoring, ONNX
Databases & Cloud: PostgreSQL, MySQL, MongoDB, Snowflake, AWS (S3, EC2, Lambda, RDS), Azure Machine Learning
Tools & Methodologies: Git, GitHub, GitLab, Jenkins, Linux, Jupyter Notebook, Jira, Confluence, Agile, Scrum

Experience

Berkshire Hathaway| AI/ML Engineer

Jan 2026 - Present

- Designed machine learning solutions to improve business decision-making by translating stakeholder requirements into production-ready predictive models supporting enterprise operational workflows.
- Developed scalable model training and deployment pipelines, reducing model delivery time by 35% while improving consistency across multiple production releases.
- Optimized inference workflows, decreasing prediction response time by 40% and enabling reliable real-time recommendations for more than 150 internal business users.
- Collaborated with cross-functional teams to validate model performance, resolve production issues, and ensure accurate prediction results through structured testing and continuous monitoring.
- Documented model development standards, evaluation procedures, and deployment guidelines to improve knowledge sharing and simplify future maintenance activities across engineering teams.

Bosch| ML Engineer

Jan 2022 - Jul 2024

- Built machine learning models to enhance manufacturing process quality by analyzing operational data and supporting predictive decision-making across multiple production environments.
- Implemented automated data preparation workflows, reducing manual processing effort by 55% while improving the consistency and reliability of training datasets.
- Improved model accuracy by 18% through systematic feature engineering, validation strategies, and continuous refinement using historical production data.
- Integrated production-ready machine learning solutions into existing business applications, enabling seamless consumption of prediction results by operational teams.
- Monitored deployed models to identify performance degradation, ensuring reliable predictions through periodic retraining and continuous evaluation of business outcomes.
- Coordinated with product owners, data engineers, and quality teams to gather requirements, validate expected outcomes, and deliver machine learning capabilities aligned with business objectives.
- Delivered multiple production model releases supporting over 200 daily operational users while maintaining high reliability and minimizing deployment-related issues.

Education

Master of Science in Computer Science

Aug 2024 - May 2026

University of Houston, Houston, TX, USA

Certifications

- Generative AI and LLMOps: Building Blocks and Applications — Coursera
- Machine Learning Specialization — Coursera (Andrew Ng)
- Introduction to Large Language Models — Google Cloud Skills Boost

Projects

Intelligent Document Question Answering System | University of Houston

- Developed an AI-powered RAG application using Python, LangChain, OpenAI APIs, and FAISS to enable intelligent document search, semantic retrieval, and context-aware question answering from uploaded PDF documents.
- Designed scalable FastAPI REST APIs, integrated PostgreSQL for metadata management, and optimized vector embeddings and retrieval performance to improve application scalability, response accuracy, and overall user experience.
- Implemented document chunking, prompt engineering, and embedding optimization techniques while evaluating retrieval quality to improve answer relevance, reduce response latency, and enhance the overall user experience.

Customer Churn Prediction System | University of Houston

- Built an end-to-end machine learning solution using Python, Scikit-learn, XGBoost, Pandas, and NumPy to predict customer churn through data preprocessing, feature engineering, model training, and predictive analytics, enabling proactive customer retention strategies.
- Optimized model performance through hyperparameter tuning, cross-validation, and comprehensive model evaluation while integrating PostgreSQL for efficient data management, improving prediction reliability and delivering actionable business insights for decision-making.